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S24CSEU2526

LAB – 6

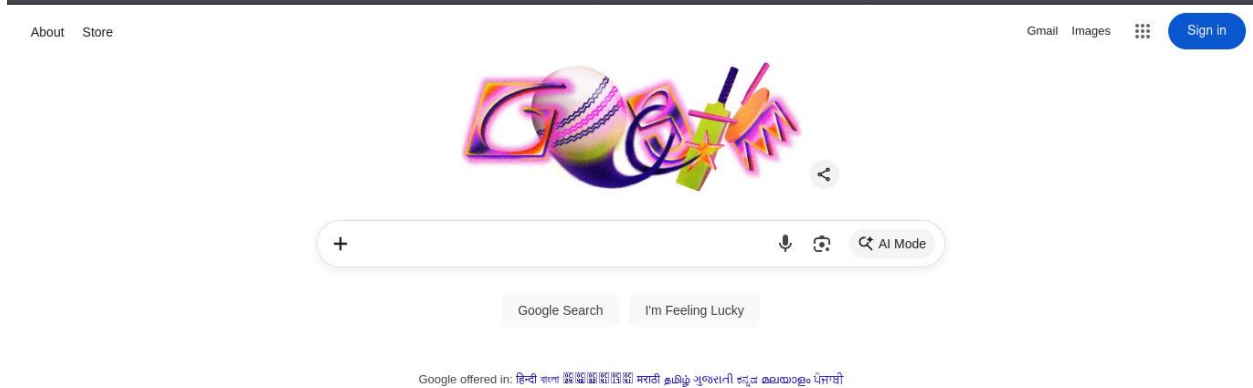
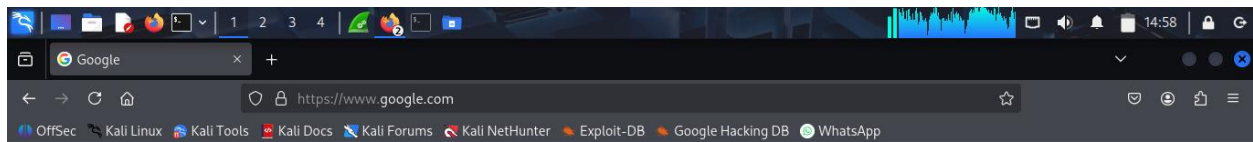
Experiment – 1

The image shows a Wireshark network traffic capture window. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains various icons for packet capture and analysis. The display filter bar shows "Apply a display filter ... <Ctrl-/>". The packet list pane shows a list of captured packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The packet details pane shows the selected packet's structure, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
2047	38.556840688	10.0.2.15	34.107.243.93	TCP	54	37196 → 443 [ACK] Seq=65 Ack=2 Win=9344 Len=0
2048	33.279980130	PCSSystemtec_e5:52:...	52:55:0a:00:02:03	ARP	42	Who has 10.0.2.3? Tell 10.0.2.15
2049	33.280550091	52:55:0a:00:02:03	PCSSystemtec_e5:52:...	ARP	64	10.0.2.3 is at 52:55:0a:00:02:03
2050	35.731925918	10.0.2.15	57.144.15.32	TCP	93	39508 → 5222 [PSH, ACK] Seq=1 Ack=1 Win=65535 Len=39
2051	35.732521631	57.144.15.32	10.0.2.15	TCP	60	5222 → 39508 [ACK] Seq=1 Ack=40 Win=65535 Len=0
2052	35.822097101	57.144.15.32	10.0.2.15	TCP	93	5222 → 39508 [PSH, ACK] Seq=1 Ack=40 Win=65535 Len=39
2053	35.822128331	10.0.2.15	57.144.15.32	TCP	54	39508 → 5222 [ACK] Seq=40 Ack=40 Win=65535 Len=0
2054	38.528613204	10.0.2.15	23.15.147.118	TCP	54	[TCP Keep-Alive] 55536 → 80 [ACK] Seq=431 Ack=727 Win=63514 Len=0
2055	38.528919038	23.15.147.118	10.0.2.15	TCP	60	[TCP Keep-Alive ACK] 80 → 55536 [ACK] Seq=727 Ack=432 Win=65535 Len=0
2056	38.752208875	10.0.2.15	23.15.147.118	TCP	54	[TCP Keep-Alive] 55546 → 80 [ACK] Seq=862 Ack=1453 Win=62788 Len=0
2057	38.752495654	23.15.147.118	10.0.2.15	TCP	60	[TCP Keep-Alive ACK] 80 → 55546 [ACK] Seq=1453 Ack=863 Win=65535 Len=0

Frame 1: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface eth0
Ethernet II, Src: PCSSystemtec_e5:52:33 (08:00:27:e5:52:33), Dst: 52:55:0a:00:02:03
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 172.66.2.5
Transmission Control Protocol, Src Port: 36484, Dst Port: 80, Seq: 1, Ack: 1453, Win: 65535, Len: 0

eth0: <live capture in progress> Packets: 2057 Profile: Default

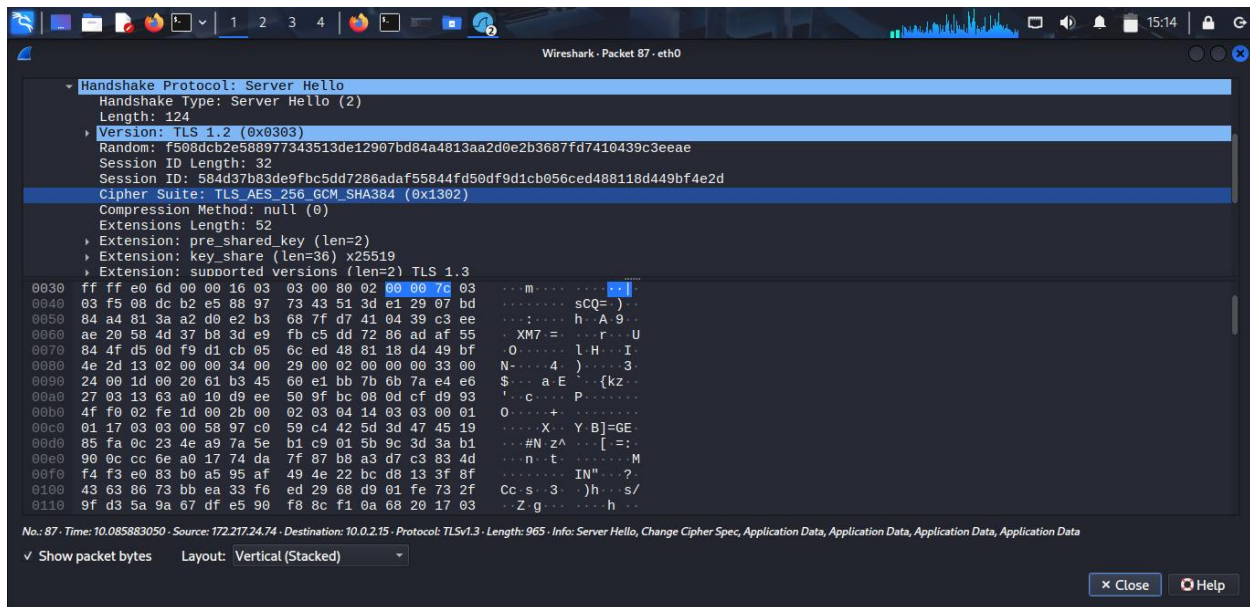


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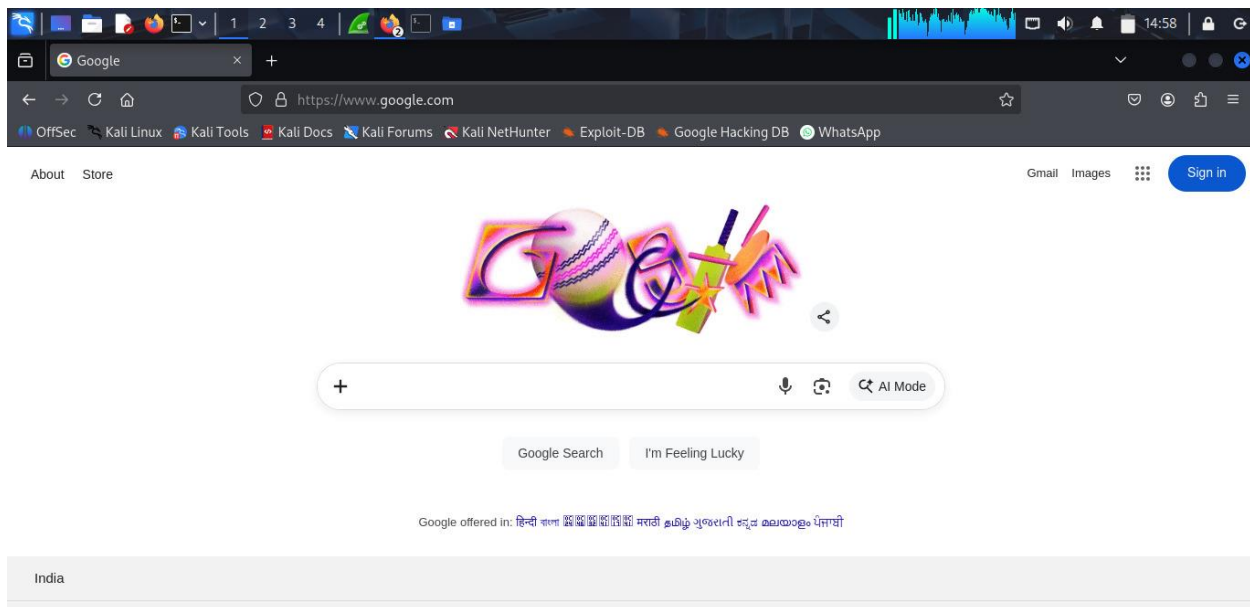
No.	Time	Source	Destination	Protocol	Length	Info
29	0.612136292	57.144.15.32	10.0.2.15	TLSv1.2	4374	Application Data, Application Data
31	0.612590839	57.144.15.32	10.0.2.15	TLSv1.2	8694	Application Data, Application Data
33	0.612775049	57.144.15.32	10.0.2.15	TLSv1.2	2934	Application Data
35	0.613094660	57.144.15.32	10.0.2.15	TLSv1.2	4374	Application Data, Application Data
37	0.613314617	57.144.15.32	10.0.2.15	TLSv1.2	3564	Application Data
39	0.618936182	57.144.15.32	10.0.2.15	TLSv1.2	2202	Application Data, Application Data
41	0.945736075	10.0.2.15	57.144.15.32	TLSv1.2	147	Application Data
43	1.342107343	57.144.15.32	10.0.2.15	TLSv1.2	186	Application Data
45	2.406802303	10.0.2.15	57.144.15.32	TLSv1.2	93	Application Data
46	2.407108134	10.0.2.15	57.144.15.32	TLSv1.2	78	Application Data
51	2.525531704	57.144.15.32	10.0.2.15	TLSv1.2	78	Application Data
70	10.012224134	10.0.2.15	57.144.41.32	QUIC	1399	0-RTT, DCID=807061fed4108c13, SCID=dd4e52
72	10.056010809	57.144.41.32	10.0.2.15	QUIC	1274	Initial, DCID=dd4e52, SCID=8000467613ecce57, PKN: 5724525, CRYPTO,
79	10.059487641	10.0.2.15	172.217.24.74	TLSv1.3	1364	Client Hello (SNI=safebrowsing.googleapis.com)
81	10.060007488	10.0.2.15	172.217.24.74	TLSv1.3	60	Change Cipher Spec
82	10.060033262	10.0.2.15	172.217.24.74	TLSv1.3	610	Application Data
87	10.085883050	172.217.24.74	10.0.2.15	TLSv1.3	965	Server Hello, Change Cipher Spec, Application Data, Application Data
92	10.087342486	10.0.2.15	172.217.24.74	TLSv1.3	154	Application Data, Application Data
94	10.088153571	10.0.2.15	172.217.24.74	TLSv1.3	85	Application Data
96	10.195241243	172.217.24.74	10.0.2.15	TLSv1.3	635	Application Data, Application Data, Application Data, Application
97	10.195628824	10.0.2.15	172.217.24.74	TLSv1.3	93	Application Data
98	10.195812255	10.0.2.15	172.217.24.74	TLSv1.3	78	Application Data
106	10.227731309	10.0.2.15	57.144.41.32	QUIC	1399	0-RTT, DCID=f3b12e7410ba801a, SCID=72aa30
107	10.248156177	57.144.41.32	10.0.2.15	QUIC	1274	Initial, DCID=72aa30, SCID=9f0057800c787ed6, PKN: 7225429, CRYPTO,
113	10.258266178	10.0.2.15	57.144.15.32	TLSv1.2	137	Application Data

Transport Layer Security: Protocol

Packets: 1055 - Displayed: 240 (22.7%) - Dropped: 0 (0.0%) - Profile: Default



Experiment – 2



Wireshark interface showing network traffic capture on eth0. The packet list displays 25 packets, including UDP, TCP, and TLSv1.2 traffic. The packet details pane shows the structure of the selected packet (Frame 1: 855 bytes on wire).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.0.2.15	142.251.43.4	UDP	855	46987 → 443 Len=813
2	0.064924762	142.251.43.4	10.0.2.15	UDP	74	443 → 46987 Len=32
3	0.065558357	10.0.2.15	142.251.43.4	UDP	74	46987 → 443 Len=32
4	0.137656442	142.251.43.4	10.0.2.15	UDP	408	443 → 46987 Len=366
5	0.142705593	142.251.43.4	10.0.2.15	UDP	67	443 → 46987 Len=25
6	0.147119648	10.0.2.15	142.251.43.4	UDP	80	46987 → 443 Len=38
7	0.207170794	142.251.43.4	10.0.2.15	UDP	71	443 → 46987 Len=29
8	8.394099899	10.0.2.15	57.144.243.32	TLSv1.2	124	Application Data
9	8.395304859	57.144.243.32	10.0.2.15	TCP	60	443 → 58246 [ACK] Seq=1 Ack=71 Win=65535 Len=0
10	8.643421719	57.144.243.32	10.0.2.15	TLSv1.2	126	Application Data
11	8.643581554	10.0.2.15	57.144.243.32	TCP	54	58246 → 443 [ACK] Seq=71 Ack=73 Win=61920 Len=0
12	12.624136871	10.0.2.15	34.107.243.93	TLSv1.2	84	Application Data
13	12.625321456	34.107.243.93	10.0.2.15	TCP	60	443 → 43798 [ACK] Seq=1 Ack=31 Win=65535 Len=0
14	17.642595324	10.0.2.15	10.0.2.3	DNS	88	Standard query 0xa79f A contile.services.mozilla.com
15	17.763267277	10.0.2.3	10.0.2.15	DNS	140	Standard query response 0xa79f A contile.services.mozilla.com CNAM
16	17.767959252	10.0.2.15	199.232.113.91	TCP	74	53338 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2
17	17.900436170	199.232.113.91	10.0.2.15	TCP	60	443 → 53338 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
18	17.900646476	10.0.2.15	199.232.113.91	TCP	54	53338 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
19	17.903946945	10.0.2.15	199.232.113.91	TLSv1.3	1177	Client Hello (SNI=contile.services.mozilla.com)
20	17.904339612	10.0.2.15	199.232.113.91	TLSv1.3	60	Change Cipher Spec
21	17.904542994	199.232.113.91	10.0.2.15	TCP	60	443 → 53338 [ACK] Seq=1 Ack=1124 Win=65535 Len=0
22	17.904794773	199.232.113.91	10.0.2.15	TCP	60	443 → 53338 [ACK] Seq=1 Ack=1130 Win=65535 Len=0
23	17.904820987	10.0.2.15	199.232.113.91	TLSv1.3	322	Application Data
24	17.905283388	199.232.113.91	10.0.2.15	TCP	60	443 → 53338 [ACK] Seq=1 Ack=1398 Win=65535 Len=0
25	18.040041030	199.232.113.91	10.0.2.15	TLSv1.3	515	Server Hello, Change Cipher Spec, Application Data

Frame 1: 855 bytes on wire (6840 bits) 855 bytes captured (6840 bits) on eth0
Packets: 47 - Dropped: 0 (0.0%) Profile: Default